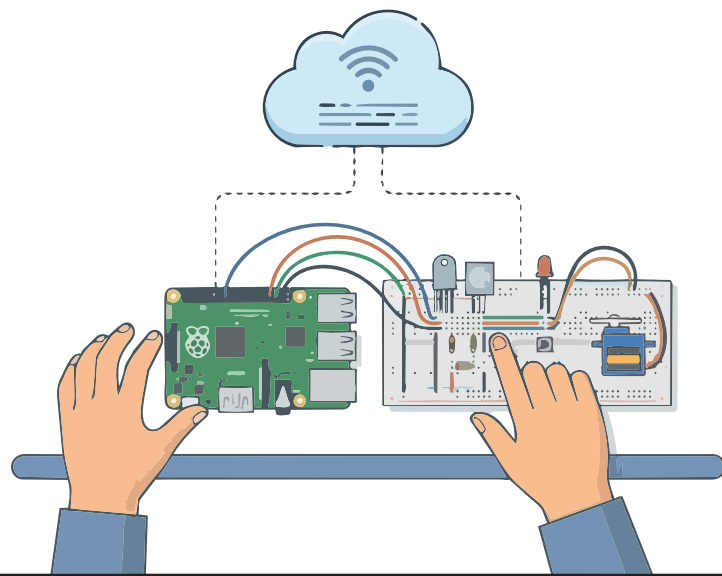


Interested in IoT?

Play with Raspberry PI

The Raspberry PI Pico board is excellent for building a range of IoT-based applications.



```
RUN wget https://github.com/micropython/micropython/archive/
refs/heads/master.zip \
  && unzip master.zip \
  && cp micropython-master/tools/pyboard.py /usr/local/bin/
pyboard \
  && rm -rf master.zip micropython-master \
  && pip install pyserial \
  && mkdir /etc/rpipico
```

```
WORKDIR /etc/rpipico
ENTRYPOINT ["pyboard"]
CMD ["-h"]
```

If you execute the command:

```
docker run --rm --device=/dev/ttyACM0:/dev/ttyACM0 pyboard
```

...it dumps a help screen to show all that you can do with it, as shown in Figure 1. Please ensure you are not connected to your RPI Pico board through Thonny, and then execute the following command:

```
docker run --rm --device=/dev/ttyACM0:/dev/ttyACM0 pyboard -c
'from machine import Pin;from time import sleep;p=Pin("LED",
Pin.OUT);p.toggle();sleep(3);p.toggle()'
```

```
usage: pyboard [-h] [-d DEVICE] [-b BAUDRATE] [-u USER] [-p PASSWORD]
               [-c COMMAND] [-w WAIT] [--soft-reset] [--no-soft-reset]
               [--follow] [--no-follow] [--exclusive] [--no-exclusive] [-f]
               [files ...]

Run scripts on the pyboard.

positional arguments:
  files                input files

options:
  -h, --help            show this help message and exit
  -d, --device DEVICE  the serial device or the IP address of the pyboard
  -b, --baudrate BAUDRATE
                        the baud rate of the serial device
  -u, --user USER      the telnet login username
  -p, --password PASSWORD
                        the telnet login password
  -c, --command COMMAND
                        program passed in as string
  -w, --wait WAIT       seconds to wait for USB connected board to become
                        available
  --soft-reset          Whether to perform a soft reset when connecting to the
                        board [default]
  --no-soft-reset       Whether to skip a soft reset when connecting to the
                        board
  --follow              follow the output after running the scripts [default
                        if no scripts given]
  --no-follow           do not follow the output after running the scripts
  --exclusive           Open the serial device for exclusive access [default]
  --no-exclusive        do not open the serial device for exclusive access
  -f, --filesystem     perform a filesystem action: cp local :device | cp
                        :device local | cat path | ls [path] | rm path | mkdir
                        path | rmdir path
```

Figure 1: Pyboard help screen

Our IoT journey started with the basics of the Raspberry Pi in an article published in the January 2025 edition of OSFY (<https://www.opensourceforu.com/2025/02/raspberry-pi-pico-the-wonder-board-for-iot-applications/>). Let's move further on this journey by using more tools and external sensors that interface with the Pi Pico boards.

Command line tools

For the command line and automation Ninjas, the MicroPython project provides tools like *pyboard* and *mpremote* to interact with the RPI Pico boards. We'll experiment with both to drive these boards through their respective Docker containers. A Dockerfile_Pyboard and execute command is given below:

```
docker build -f Dockerfile_Pyboard -t pyboard to create
pyboard docker image.
```

```
FROM python:3.14-alpine
SHELL ["/bin/ash", "-o", "pipefail", "-c"]
WORKDIR /tmp
```